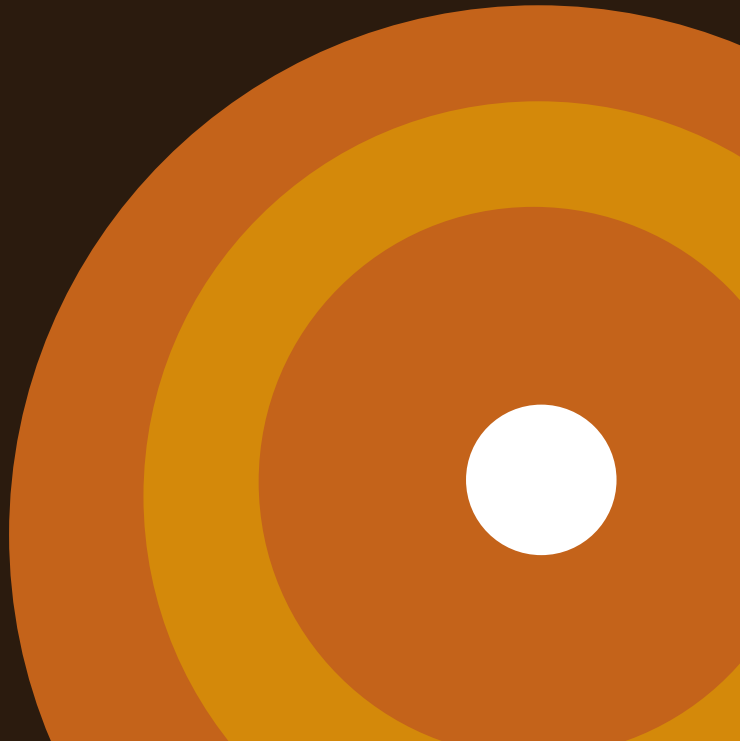
Soumeng Chea

UX Researcher | Human Factors | Product Strategy

Designing products that transform complex systems into intuitive human experiences.



SOUMENG CHEA | UX RESEARCHER | HUMAN FACTORS | PRODUCT STRATEGY | USER EMPATHY ENGINEER

About Me



I uncover how people think so technology can work better for them.

Three Pillars

1

Human Factors

Cognition, behavior & decision making

2

UX Research

Insights into actionable recs

3

Product Strategy

Connecting needs with biz goals

Human Factors → **Research** → **Product Decisions** → **Better User Outcomes**

Case Study Portfolio

01

AI Healthcare Comprehension Study

AI in Healthcare | UX Research | Usability Testing

02

Smart OBDS Device Design

Sanofi Design Challenge 2025 | Human Factors | Product Design

03

HydraTrim Smart Shaver

L'Oréal Brandstorm | Product Strategy

04

Whisk & Whisper Voice Cooking Assistant

Bentley University | Conversational UI

01

AI Healthcare Comprehension Study

Evaluating how patients interpret AI-generated appointment summaries — and what builds trust in clinical AI tools.

Overview

The Challenge

Patients often leave appointments confused by complex medical notes. AI tools aim to simplify health information — but does simplification guarantee true understanding? For Client A, patient trust directly determines adoption of AI-enabled tools.

Role

- Co-Lead
- UX Analyst
- Moderation
- Experience Mapper

Methods

5 remote moderated sessions | UserTesting platform | Think-aloud protocol
Pre/post comprehension ratings | Likert scale | Task performance metrics

Key Findings — What Worked

1 Comprehension

Patiently.ai measurably improves comprehension across all literacy levels. Post-use, 4 of 5 participants rated trust at 7/7.

> *"I would say the simplified note is well done. It's giving me a good breakdown and explaining things without getting too detailed."* – P1

2 Trust Features

No login required, PHI/PII auto-redacted, MHRA-registered status, and clear AI disclaimers were table-stakes trust signals.

> *"It exceeded my expectations. There are a lot of customizable features. I kind of expected you would just paste and get a summary."* – P2

3 Engagement

Question builder, reading-level customization, and original-vs-summary comparison created high-value 'aha' moments.

> *"I'm not getting medical advice, I'm just getting a better explanation of what it means."* – P4

Usability Barriers Identified

Input Limitations

5,000-char limit, no image support, and difficulty combining multiple records created unexpected friction and reduced trust.

Trust Degradation

'87% medically accurate' read as '1 in 10 could be wrong.' 'Try for Free' and inconsistent privacy language eroded confidence.

Question Generator

Highest frustration rating (avg 3.25/7). Overwhelming volume, generic questions, not personalized or editable in-app.

Calendar Workflow

Double failure: feature wasn't where expected, and didn't pre-populate appointment details. Most difficult task.

Research-Driven Recommendations

Core Opportunity

Integrate AI-powered health literacy directly into Client A's patient portal (Marlo) to leverage innate clinical trust, which is something public-facing AI tools cannot replicate. Patients are already turning to general-purpose AI tools like ChatGPT without clinical grounding.

Portal Integration

Add 'Summarize Appointment Notes' to patient dashboard: removes 'Try for Free' concern, connects to health records, builds trust.

Automated Note Import

Pull notes directly from patient records to eliminate 5,000-char friction and multi-source input problems.

Context-Aware Chatbot

Replace static question list with a personalized chatbot: relevant, not overwhelming, familiar UX pattern.

Smart Calendar

Auto-fill appointment data into calendar with one-click personal calendar add: seamless AI-forward experience.

02

Smart OBDS Device Design

Designing the next-generation wearable On-Body Drug Delivery System for Sanofi's 2025 Device Innovation Challenge.

Overview & The Problem

The Design Challenge

How might we create a connected, large-volume OBDS that patients can trust to stay secure and fit naturally into their daily lives?

1 in 3

therapies not taken as recommended

26%

outcomes gap: high vs. low adherence

1 in 5

experience injection anxiety

Gap Analysis — Current Devices Lack:

Support	Comfort for >10 mL volumes subcutaneously
Accessibility	Limited options for visual/motor/cognitive impairments
Discretion	Rigid medical forms not wearable for long duration
Flexibility	High environmental impact; form factors induce fear

User Research — Personas & Key Insights

Maya

48 yrs

High school teacher & avid hiker. HER2+ breast cancer. Starting self-injection after hospital IV infusions.

Axel

31 yrs

Marathon runner & Marketing Manager. Type 1 Diabetes. Data-driven, tech-savvy, constantly optimizing.

Chloe

29 yrs

Architect on maternity leave. Gestational Diabetes Mellitus (28 wks). Unexpected complexity added to pregnancy.

Key Insights from Affinity Mapping

Adhesive

Users want flexible, form-hugging adhesives — but worry about adhesion during sweat and showers.

Discretion

Device must be discreet in form AND feedback. Tubing, alerts, and medical look all create stigma.

Technology

Visual feedback on both phone and device is desired, but users resist phone-only control.

Design Solution — EasePatch

Care that's easy to wear

Advanced Usability

Pre-filled, sterile patch with automatic needle insertion/retraction. LED + haptic feedback guide users step by step (blue=ready, yellow=delivery, red=error).

Flexible Wear

Apply patch directly to skin OR use optional belt system for active users. Belt features skin-tone adaptability and antibacterial woven mesh with color-changing textile.

Technical Performance

Delivers up to 25 mL subcutaneously across multiple therapeutic areas. Worn comfortably for 5+ hours without irritation.

Pain-Free + Safety

Belt-integrated TENS technology provides up to 80% pain reduction. Real-time vital tracking (HR, BP) builds caregiver confidence.

User Journey & Impact

Receive — Attach — Confirm — Deliver — Notify — Replace

1

Patient Empowerment

Reduce pain, anxiety, and complexity. Adaptable patch-or-belt wear based on user lifestyle.

2

Innovation with Purpose

TENS numbing + vitals tracking transform medical devices into holistic health companions.

3

Future of OBDS

A discreet, patient-centered solution that doesn't just treat disease, it enhances daily living.

03

HydraTrim Smart Shaver

World's first hydration-tracking smart shaver — redefining manscaping for Gen Z through sustainability, AI, and inclusive design. L'Oréal Brandstorm Competition.

Product Overview

The Product

HydraTrim is the world's first hydration-tracking smart shaver for Gen Z men. Combining bio-impedance sensors for real-time skin moisture detection, surgical-grade blades, and an AI virtual assistant, all paired with a blade recycling program and modular replacement system.

Smart

Bio-impedance hydration sensors + AI assistant accessible 24/7

Sustainable

Blade recycling program + modular replacement + low-energy BT chip

Inclusive

Engineered for all skin types, hair textures, and shaving preferences

Price Point

$70–$100 retail, most advanced tech at moderate cost

Market Opportunity & Go-to-Market Strategy

$11.89B
Global electric razor market (2023)

$17.66B
Projected market value by 2032 (4.6% CAGR)

$15M
Marketing budget allocation for Gen Z

Marketing Finance Allocation

Influencer Marketing	40%
Social Media Ads	30%
Loyalty Program	20%
E-Commerce Integration	10%

Competitive Positioning

HydraTrim positions in the 'Innovative / Moderate Price' quadrant, capturing half the market by offering the most technologically advanced solution at an accessible price point versus Braun, Philips, and Panasonic.

KPI Milestones

0–6 mo	CTR, Units Sold, Engagement
6–18 mo	Purchase Rate, Recycling Participation
18+ mo	Market Share Growth, Clinical Validation

04

Whisk & Whisper

Designing a hands-free, voice-first conversational cooking assistant for Gen Z adults who discover recipes through social media — Bentley University HF764.

Overview & Research

The Problem

Over half of Gen Z consumers discover recipes via social media. Yet following short-form video recipes while cooking creates massive cognitive load, constant video rewinding, extracting steps hidden under visual effects, losing place mid-cook.

Whisk & Whisper bridges this gap: a hands-free conversational assistant that lets users navigate recipes entirely by voice while a visual display provides glanceable support.

Design Process

Research	User research + competitor analysis
Define	User journey mapping → design opportunities
Ideate	Feature brainstorm + MVP prioritization
Design & Dev	Conversational flow, wireframes, prototypes
Test & Iterate	5 usability sessions + iterative refinement

Voice Bot Design & Key Features

Meet Chef

Chef is a 24-year-old reliable 'cooking friend' who has been in the kitchen since a young age. Warm, supportive, resourceful, and organized, where Chef adapts to the user's pace and communicates in a clear, energetic, friendly tone.

Intent Map

START	Ready to begin — hear first step
NEXT	Step complete — move forward
REPEAT	Didn't catch it — hear step again
PREVIOUS	Missed a step — go back
TIP	Want to know how to do this better

Core Features

Voice-Controlled Navigation

Hands-free step navigation via voice commands. Mic auto-pauses during TTS playback to prevent looping.

Social Media Recipe Import

Import recipes directly from Instagram/TikTok. Eliminates rewinding — one-tap to voice-guided cooking.

Visual Guidance

Glowing mic indicator shows listening/speaking states. Step progress displayed for glanceable reference.

Tech Stack: HTML/CSS/JS + Node.js + Express · Web Speech API (STT) · OpenAI.fm TTS (gpt-4o-mini-tts)

User Testing & Iterative Improvements

5
moderated sessions (ages 21–35)

2
tasks per session (Figma + Voice prototype)

2.5/5
avg satisfaction — clear room to improve

What Went Wrong

- Assistant heard itself → unintended 'next' loops
- "Clarify" / "Repeat" referred to wrong step
- System lacked step-state memory
- Instructions static — didn't guide next action
- No clear completion signal

How We Fixed It

- Mic auto-pauses while TTS speaks; wake-word ready
- Each step now state-tracked (currentStep, lastCommand)
- Assistant references only the active step
- Ingredient confirmation flow added (spoken + visual)
- Completion cue: 'Congratulations, you finished!'

Thank You

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Four case studies.

One throughline: Designing products that transform complex systems into intuitive human experiences.

